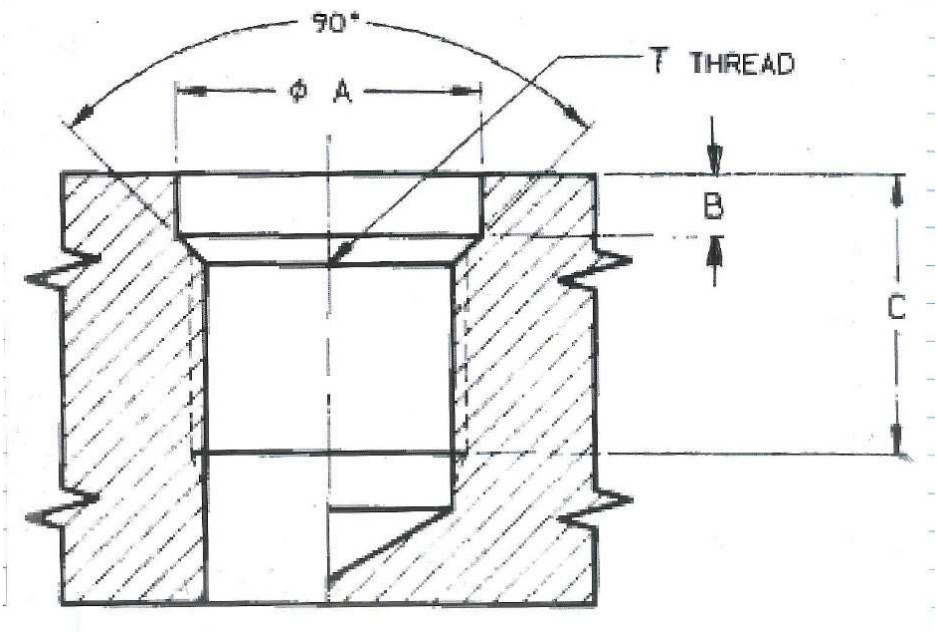
INCH-POUND

MS51994B
31 OCT 2019
SUPERSEDING
MS51994A
12 FEB 1988

DETAILED SPECIFICATION SHEET

HOLE PREPARATION – RING LOCKED INSERTS AND STUDS STANDARD DIMENSIONS FOR



REQUIREMENTS:

1. Diameter ‘A’ and minor diameter of the thread to be concentric within .003 FIM.

2. Axis of the hole to be normal to the entry surface or provide spotface when required.

3. Thread as shown per SAE-AS8879 is preferred. Alternate thread per MIL-S-7742 Class 3B or National Class 3 is acceptable when specified by design activity.

4. Unless otherwise specified, machined surfaces to be 125 microinches in accordance with ASME B46.1.

5. Remove all burrs and sharp edges.

6. Tolerances for linear dimensions +/- .005”, angular dimensions +/- 5°.

AMSC N/A FSC 53GP

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NOTES:

1. Dimensions are in inches.

2. Procedure and installation of locked ring insets, studs and lockrings, see MS51995.

3. General use: For locked ring inserts (Table I) and studs (Table II).

4. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.

5. Referenced Government documents are available on the ASSIST Web Site at https://assist.dla.mil or the ASSIST Quick search web site at https://quicksearch.dla.mil.

6.1 Referenced non-government documents are available from Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, Pa 15096-0001. http://www.sae.org/ or ASME, Two Park Avenue, New York, NY 10016-5990. http://www.asme.org/.
6.2 Changes from previous issue: Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensive changes.

THIS IS A DESIGN STANDARD. NOT TO BE USED AS A PART NUMBER.

TABLE I – INSERT MS51991

INSERT DASH NUMBER MS51991	T THREAD CLASS 3B (SEE REQT 3)	ϕA + .003 - .000		B + .030 - .000	C MIN FULL THREAD DEPTH
		AL	MAG		
102 & 202	0.1900-24UNJC	0.250	0.258	0.094	0.285
103 & 203	0.2160-24UNJC	0.281	0.291	0.109	0.345
104 & 204	0.2500-20UNJC	0.328	0.337	0.109	0.415
105 & 205	0.3125-18UNJC	0.375	0.381	0.141	0.475
106 & 206	0.3750-16UNJC	0.437	0.448	0.156	0.535
107 & 207	0.5000-13UNJC	0.515	0.526	0.172	0.595
108 & 208	0.5625-12UNJC	0.593	0.601	0.172	0.655
109 & 209	0.6250-11UNJC	0.734	0.747	0.218	0.725
110 & 210	0.7500-10UNJC	0.875	0.888	0.203	0.785
311	0.8750-14UNJF	1.000	1.015	0.203	0.915
312.1	1.0000-12UNJF	1.125	1.139	0.203	1.035
312	1.0000-14UNJS	1.125	1.139	0.203	1.035
313	1.1250-12UNJF	1.281	1.298	0.219	1.155
314	1.2500-12UNJF	1.406	1.427	0.219	1.285
315	1.3750-12UNJF	1.526	1.579	0.219	1.415

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TABLE II - STUD, MS51551 & MS51989

STUD DASH NUMBER		T THREAD CLASS 3B (SEE REQT 3)	*A +.003 -.000		B +.030 -.000	MIN FULL THREAD DEPTH			
						MS51551			MS51989 DASH 100 THRU 400 SERIES
MS51551	MS51989		AL	MAG		DASH 100 & 200 SERIES	DASH 100A & 200A SERIES	DASH 100B & 200B SERIES	
---	102	0.1640-32UNJC	0.250	0.258	0.094	---	---	---	0.285
---	302								
---	202	0.1640-36UNJF							
---	402								
103	103	.1900-24UNJC	0.281	0.291	0.109	0.415	0.535	0.655	0.415
103A, 103B	303								
203	203	.1900-32UNJF							
203A, 203B	403								
104	104	.2500-20UNJC	0.328	0.337	0.109	0.475	0.595	0.725	0.4775
104A, 104B	304								
204	204	.2500-28UNJF							
204A, 204B	404								
105	105	0.3125-18UNJC	0.375	0.381	0.141	0.595	0.725	0.845	0.595
105A, 105B	305								
205	205	0.3125-24UNJF							
205A, 205B	405								
106	106	0.3750-16UNJC	0.437	0.448	0.156	0.725	0.915	1.095	0.725
106A, 106B	306								
206	206	0.3750-24UNJF							
206A, 206B	406								
107	107	0.4375-14UNJC	0.515	0.526	0.172	0.785	0.975	1.155	0.785
107A, 107B	307								
207	207	0.4375-20UNJF							
207A, 207B	407								
108	108	0.5000-13UNJC	0.593	0.601	0.172	0.845	1.035	1.225	0.845
108A, 108B	308								
208	208	0.5000-20UNJF							
208A, 208B	408								
109	109	0.56250-12UNJC	0.734	0.747	0.188	0.915	1.095	1.345	0.915
109A, 109B	309								
209	209	0.5625-18UNJF							
209A, 209B	409								
---	110	0.6250-11UNJC	0.734	0.747	0.188	---	---	---	1.035
---	310								
---	210	0.6250-18UNJF							
---	410								
---	111	0.7500-10UNJC	0.875	0.888	0.203	---	---	---	1.155
---	311								
---	211	0.7500-16UNJF							
---	411								
---	112	0.8750-9UNJC	1.000	1.015	0.203	---	---	---	1.345
---	312								
---	212	0.8750-14UNJF							
---	412								

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MILITARY INTERESTS

Custodians:

Army - AR
Navy - AS

Preparing activity:

DLA-IS

(Project No. 53GP-2019-001)

NOTE: The activities listed above were interested in this document asof the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above usingthe ASSIST Online database at https://assist.dla.mil.

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